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Claim-Centric Multi-Document Scientific Summarization with Provenance Verification

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ABSTRACT

Synthesizing literature reviews across conflicting research papers requires isolating individual claims and mapping evidentiary agreements and contradictions. We develop ClaimSynth, a multi-document summarization system that extracts atomic scientific claims and organizes them into comparative evidence matrices. The model uses natural language inference to label claim relations as supporting, contradicting, or extending prior findings before generating summary prose. In human evaluations with doctoral researchers, ClaimSynth generated summaries rated 42% more coherent and 3.8x more faithful than standard GPT-4 summaries. However, nuanced scientific hedging and conditional experimental constraints are occasionally flattened into over-simplified binary contradictions.

Keywords— Natural Language Processing, Summarization, Multi-Document, Scientific Literature

INTRODUCTION

Automatic literature surveys often blur distinct methodological assumptions, synthesizing misleading consensus statements. ClaimSynth treats scientific texts as networks of attributed claims that must be verified and compared before synthesis.

METHODOLOGY

We employ an open-information extraction pipeline fine-tuned on academic abstracts to identify (subject, predicate, finding) triples. A cross-document NLI model clusters claims by topic and flags direct contradictions across experimental results.

RESULTS

In trials across 100 literature review topics, 88% of human evaluators preferred ClaimSynth's structured contradiction-aware summaries over unconstrained generative models, citing superior transparency.

LIMITATIONS

Scientific nuance is occasionally lost; subtle qualifications (e.g., 'under high temperature and specific pH') can be omitted by the extraction pipeline, producing false contradiction alerts.

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